Matthew Murden

Computer Science Student

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Education

University of Birmingham, BSc in Computer Science with Year in Industry — Sept 2024 – June 2028

- Year 2: 65.7% (2:1)
- Key Modules: Software Engineering (78%), Functional Programming (78%), Security & Networks (73%)

Hymers College, A-Levels — June 2024

- Mathematics (A), Physics (A), Economics (B); Financial Studies AS-Level (A)

Experience

Intern Firmware Engineer, Honeywell Technologies – Industrial Placement — July 2026 – July 2027

Working on a next-generation, safety-critical ESSER fire alarm panel

- Engineered a Python tool to transfer files to embedded Linux CPUs over serial with base64 encoding and MD5 checks, cutting transfers from 20 minutes to 10 seconds.
- Diagnosed a failing internal tool and shipped a standalone fix, now used across my team.
- Currently driving the team's test automation with hardware-in-the-loop rigs (Arduino relays, photodiodes, virtual HMI), working to turn weeks of manual testing into unattended overnight runs.

Web & Cyber Security Lead, TM Solicitors – Part-Time — June 2024 – present

- Led an authorised security audit of the firm's WordPress estate, uncovering two forgotten legacy installs and 39 CVSS-rated findings, including 9 vulnerabilities exploitable without login.
- Patched PHP2Shell (CVSS 9.2) within 2 hours of the fix's release and remediated a CVSS 9.8 plugin admin takeover the same day; both were actively exploited in the wild.
- Got a scam domain impersonating the firm taken down within 20 hours via multiple abuse channels, after the firm's own reports had failed, and hardened email and DNS security (SPF, DKIM, DMARC).

Projects

Seven-Pillar BTC Trading System | github.com/matthewmurden/btc-seven-pillar-trader — Nov 2025 – Dec 2025

- Built an automated BTCUSDT perpetual futures trading system where seven pillars each own one decision (regime detection, entry, exit, risk sizing, safety, execution, monitoring) and any pillar can veto a trade.
- Entry, exit and regime models blend hand-written rules with LightGBM classifiers on features from 1-minute candles, trade-flow delta and higher timeframes.
- Validated with walk-forward testing, a look-ahead leak test and fee/slippage stress tests.
- Runs live on the Binance demo account with limit-first execution, protective TP/SL orders and a kill switch.

Polymarket Insider Detection | github.com/matthewmurden/polymarket-insider-detection — Jan 2026 – present

- Built a live C++ pipeline streaming Polymarket trades over WebSocket and linking each one to its on-chain transaction and wallet on Polygon, measuring a ~2.2s lag between feed and settlement.
- Developed a Python anomaly scorer combining per-wallet signals into one risk score, validated against known insider cases, flagging clean control wallets at a lower rate than confirmed insiders.

Resell Profit | github.com/matthewmurden/resell-profit-ios — Sept 2025

- SwiftUI app tracking purchases, sales and live profit, with Swift Charts, CSV export, Core Data and Firebase auth.

Sorting Algorithm Visualiser | github.com/matthewmurden/sorting-visualiser — Sept 2025

- React, TypeScript and Canvas visualiser where algorithms run as step-emitting generators you can play, pause and step.

Skills

Languages: Python, C++, Java, Swift, JavaScript/TypeScript, SQL

Data & ML: pandas, polars, DuckDB, LightGBM, scikit-learn, walk-forward validation, anomaly detection

Systems & Security: Linux, embedded Linux, Git, WebSockets, test automation, HIL testing, CVSS, SPF/DKIM/DMARC